

Tejas Kadam

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Education

Vellore Institute of Technology, Vellore

2024 – 2028

B.Tech in Computer Science and Business Systems

Relevant Coursework: Data Structures, Algorithms, Machine Learning, Statistics, Database Systems

Technical Skills

Languages: Python, C++, SQL

Frameworks & Tools: Flask, FastAPI, Node.js, React, MongoDB, Git, Linux

Machine Learning: Neural Networks, Transformers, Fine-tuning, Feature Engineering, Time-Series Modeling, Hugging Face

Hardware / Systems: Raspberry Pi, Arduino, ESP32

Projects

Artemis – AI System for Financial Call Compliance Analysis

Code

- Built an AI system to analyze financial call recordings and detect compliance risks using NLP.
- Implemented audio preprocessing including speaker diarization, language detection, and speech-to-text transcription.
- Applied transformer-based models for entity extraction and intent detection in financial conversations.
- Designed a compliance scoring module to identify commitments and potential regulatory risks.

ShopTrack – Retail Sales Forecasting and Analytics Platform

Code

- Developed a time-series forecasting system to predict category-wise retail sales trends.
- Designed a modular pipeline for preprocessing, forecasting, and prediction filtering.
- Built a web dashboard with interactive visualizations and MongoDB-based data storage.

SpendWise – Personal Finance Dashboard

- Developed a web application for tracking expenses and managing spending categories.
- Built financial visualizations using D3.js to analyze spending patterns.
- Implemented a recommendation engine suggesting optimized spending based on budget constraints.

Digit Recognition using Neural Networks from Scratch

Code — Live

- Implemented a neural network from scratch in Python using NumPy to classify handwritten digits (MNIST).
- Developed forward propagation, Leaky ReLU activation, softmax output, and manual backpropagation.
- Built and deployed an interactive interface allowing users to draw digits and obtain predictions.

Reinforcement Learning Grid Visualizer

Code — Live

- Implemented a Q-learning agent to learn optimal paths in a customizable grid environment.
- Built visualizations to display state values and learning progression during training.
- Developed an interactive interface allowing users to modify RL parameters.

Gesture Recognition using Gyroscope Sensors

- Collected motion data from gyroscope sensors to train a gesture recognition model.
- Integrated the trained model with Raspberry Pi for gesture-based home automation.

Machine Learning Experience

- Fine-tuned transformer models for aspect-based sentiment analysis on text datasets.
- Implemented preprocessing pipelines and evaluation metrics for NLP classification tasks.
- Built neural networks from scratch using NumPy to understand training dynamics and optimization.

Achievements

- Finalist in two university-level hackathons among multiple competing teams.
- Solved **370+ algorithmic problems** on LeetCode focusing on data structures and algorithms.
- Secured **99.4 percentile in Physics** in JEE Main.
- Achieved **98.6 percentile in Class 10**.

Research and Learning

- Studied neural network implementation from *Make Your Own Neural Network* by Tariq Rashid.
- Exploring reinforcement learning, computational statistics, and time-series modeling.
- Practicing advanced algorithmic problem solving and dynamic programming.